Positive scalar curvature for asymptotic cones of cohomogeneity one expanders

We study the asymptotic cones of Rajan's expanding Ricci solitons on $S^2 \times \mathbb{R}^2$, with the aim of finding cones with positive scalar curvature. We will fix the size of the singular S^2 orbit and show that taking $f''(0)$ sufficiently negative makes the spherical asymptotic slope small enough to ensure this positivity.

We begin by fixing our notation, keeping Rajan's normalization so that we can use his existence and asymptotic results directly.

Definition 1 (The family and notation)**.** For $b_0 > 0$ and $f_0 < 0$, we consider the smooth soliton (g, f) satisfying $\mathrm{Ric}_g + \nabla_g^2 f + g = 0$, with

$$g = dr^2 + a(r)^2 g_{S^1} + b(r)^2 g_{S^2}, \qquad r \geq 0.$$

Here $g_{S^1} = d\theta^2$, with θ of period 2π, and g_{S^2} has sectional curvature 1. We prescribe $a(0) = 0$, $a'(0) = 1$, $b(0) = b_0$, $b'(0) = 0$, $f(0) = f'(0) = 0$, and $f''(0) = f_0$, where a extends oddly and b, f extend evenly across 0. We write $\lambda = \sqrt{-2f_0}$, $c = 1 + b_0^{-2}$, $\alpha = \lim_{r\to\infty} a'(r)$, and $\beta = \lim_{r\to\infty} b'(r)$.

Since we have chosen exactly the normalization and boundary conditions in [Raj25, §2, (2.6)–(2.7)], we can use Rajan's Theorem 5.1 to obtain completeness and his Lemmas 6.12–6.13 to conclude that $\alpha, \beta > 0$. His Theorem 6.14 identifies the asymptotic cone as $\gamma = ds^2 + s^2(\alpha^2 g_{S^1} + \beta^2 g_{S^2})$, and we will bound β because this slope alone determines its scalar curvature.

Theorem 2. *If $\lambda^2 \geq 8c$, then*

$$0 < \beta^2 \leq \frac{b_0^2 + 1}{\lambda^2 - 4c}. \tag{1}$$

Thus, for each fixed b_0 and $C > \sqrt{b_0^2 + 1}$, we have $\beta \leq C/\sqrt{-2f_0}$ once f_0 is sufficiently negative, and the corresponding asymptotic cones have strictly positive scalar curvature away from their vertices.

We will prove Theorem 2 after Lemma 3, which shows that $(b^2 + 1)/(\lambda^2 - 2f - 4c)$ decreases along the soliton. By comparing this quotient at the singular orbit and at infinity, we will obtain the bound for β, which we can then use to check the scalar curvature of γ.

Lemma 3 (A monotone quotient)**.** *If $\lambda^2 \geq 8c$, then $Q(r) = (b(r)^2 + 1)/(\lambda^2 - 2f(r) - 4c)$ strictly decreases for $r > 0$.*

Proof. We set $v = -f$ and $d = 4c$, so that the denominator of Q is $P = \lambda^2 + 2v - d$, and write $H = a'/a + 2b'/b + v'$ on $(0, \infty)$. By [Raj25, Lemmas 4.1–4.2], we have $a', b', v', v'' > 0$, and hence $b \geq b_0$ and $v \geq 0$. Rajan's equation (2.3) reads $b'' = b + [1 - (b')^2]/b - (a'/a + v')b'$, so adding $Hb' = (a'/a + v')b' + 2(b')^2/b$ gives the first equation below. For the second, his Lemma 3.1 gives $\Delta f - |\nabla f|^2 - 2f = 2f_0 = -\lambda^2$ in the $S^2 \times \mathbb{R}^2$ case. The radial volume density is proportional to ab^2, so $\Delta f = (ab^2)^{-1}(ab^2 f')' = f'' + (a'/a + 2b'/b)f'$. Substituting $f = -v$ and $|\nabla f|^2 = (f')^2$ therefore yields

$$b'' + Hb' = b + \frac{1 + (b')^2}{b}, \qquad v'' + Hv' = \lambda^2 + 2v. \tag{2}$$

Since $H \geq v'$ and $v'' > 0$, the second equation gives $(v')^2 \leq Hv' < v'' + Hv' = P + d$. Our assumption $\lambda^2 \geq 8c$ gives $P \geq \lambda^2 - d \geq d > 0$, so $(v')^2/P < 1 + d/P \leq 2$.

To show that Q cannot stop decreasing, we set $Y = b^2 + 1$, so $Y = QP$, and compute $Y' = 2bb'$ and $P' = 2v'$. Equation (2) then gives $Y'' + HY' = 2(b')^2 + 2b(b'' + Hb') = 2Y + 4(b')^2$ and $P'' + HP' = 2(v'' + Hv') = 2P + 2d$. At a critical point of Q, differentiating $Y = QP$ gives $Y' = QP'$ and $Y'' = PQ'' + QP''$, because $Q' = 0$. Consequently, $PQ'' = (Y'' + HY') - Q(P'' + HP') = 4(b')^2 - 2dQ$, while $bb' = Qv'$. Using $Q = (b^2 + 1)/P$, we can rewrite the latter identity as $(b')^2 = Q^2(v')^2/b^2 = Q(1 + b^{-2})(v')^2/P$, which gives

$$Q'' = \frac{4(b')^2 - 2dQ}{P} = \frac{2Q}{P}\left(2(1 + b^{-2})\frac{(v')^2}{P} - d\right) < \frac{2Q}{P}(4c - d) = 0.$$

Here we used $1 + b^{-2} \le c$ together with $(v')^2/P < 2$.

To check the initial sign of Q', we use oddness of a and evenness of b, v to write $a(r) = r + O(r^3)$, $b'(r) = b''(0)r + O(r^3)$, and $v'(r) = v''(0)r + O(r^3)$. Thus $a'/a = r^{-1} + O(r)$ and $H = r^{-1} + O(r)$, so $b'' + Hb' \to 2b''(0)$ as $r \downarrow 0$. Taking the limit in the first equation of (2) gives $2b''(0) = b_0 + b_0^{-1}$. It follows that $Y''(0) = 2b_0 b''(0) = b_0^2 + 1 = Y(0)$, while $P''(0) = 2v''(0) = -2f_0 = \lambda^2$ and $Y'(0) = P'(0) = 0$. The quotient rule now gives $Q'(0) = 0$ and

$$Q''(0) = \frac{Y''(0)}{P(0)} - \frac{Y(0)P''(0)}{P(0)^2} = \frac{b_0^2 + 1}{\lambda^2 - d} - \frac{(b_0^2 + 1)\lambda^2}{(\lambda^2 - d)^2} = -\frac{d(b_0^2 + 1)}{(\lambda^2 - d)^2} < 0.$$

Thus Q' is negative for small positive r. If Q' had a first zero, we would have $Q'' \ge 0$ there, contradicting our calculation at a critical point of Q. □

To finish Theorem 2, we will use Lemma 3 to bound the limit of $Q = (b^2 + 1)/(\lambda^2 - 2f - 4c)$ by $Q(0)$, then identify that limit with β^2 and check the scalar curvature of γ.

Proof of Theorem 2. For the solution in Definition 1, we can apply [Raj25, Lemmas 6.10 and 6.13] to obtain $f'(r) = -r + O(1)$ and $b(r)/r \to \beta$. After integrating the estimate for f', we have $f(r) = -r^2/2 + O(r)$ for each fixed soliton. Returning to $Y = b^2 + 1$ and $P = \lambda^2 - 2f - 4c$, we have $P/r^2 = 1 + O(r^{-1}) \to 1$ and $Y/r^2 = (b/r)^2 + r^{-2} \to \beta^2$. Lemma 3 therefore gives $\beta^2 = \lim_{r\to\infty}(Y/r^2)/(P/r^2) = \lim_{r\to\infty} Q(r) \le Q(0)$, which is (1). Multiplying that bound by λ^2, we obtain $(\lambda\beta)^2 \le (b_0^2 + 1)/(1 - 4c/\lambda^2)$. For fixed b_0, the right side tends to $b_0^2 + 1 < C^2$, which proves the estimate for every $C > \sqrt{b_0^2 + 1}$.

We next compute the scalar curvature of γ using the sectional curvature formulas in [Raj25, Appendix A]. Those formulas apply to any metric of the form in Definition 1, so we substitute the cone profiles $a(s) = \alpha s$ and $b(s) = \beta s$. In an orthonormal frame adapted to the radial, circle, and sphere directions, the three radial sectional curvatures are $-a''/a = 0$ and $-b''/b = 0$ (the latter twice). The two circle–sphere curvatures are $-a'b'/(ab) = -\alpha\beta/(\alpha\beta s^2) = -s^{-2}$, while the sphere curvature is $[1 - (b')^2]/b^2 = (\beta^{-2} - 1)/s^2$. Since scalar curvature is twice the sum over the six coordinate planes, we obtain

$$R_\gamma(s) = 2\left(-\frac{1}{s^2} - \frac{1}{s^2} + \frac{\beta^{-2} - 1}{s^2}\right) = \frac{2/\beta^2 - 6}{s^2}. \tag{3}$$

We thus need $\beta < 1/\sqrt{3}$, which follows from (1) as soon as $\lambda^2 > 8c$ and $\lambda^2 > 4c + 3(b_0^2 + 1)$. □

We have therefore proved positivity for every fixed b_0 once f_0 is sufficiently negative, and we can use (1) and (3) to check the concrete choice $b_0 = 1$, $f_0 = -100$.

Remark 4 (An explicit example)**.** If we choose $b_0 = 1$ and $f_0 = -100$, then $\lambda^2 = 200$, $c = 2$, and (1)–(3) give $\beta^2 \leq 2/(200 - 8) = 1/96$ and $R_\gamma(s) \geq (2 \cdot 96 - 6)/s^2 = 186/s^2$ for every $s > 0$.

We thus obtain a concrete soliton on $S^2 \times \mathbb{R}^2$ whose asymptotic cone has positive scalar curvature, with a curvature bound determined by its initial data.

References

[Raj25] Abishek Rajan, *Cohomogeneity One Expanding Ricci Solitons and the Expander Degree*, 2025. arXiv:2510.15192v2. All numbered citations refer to version 2.